Omer Kurkutlu

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Ph.D. Candidate in Electrical and Computer Engineering

Location: Chicago, IL, USA

Research Profile

Ph.D. candidate in Electrical and Computer Engineering specializing in robot learning, Physical AI, and autonomous systems. Research focuses on deep reinforcement learning, vision-language-action models, vision-based navigation, sim-to-real, and resource-constrained robotics, with hands-on experience in NVIDIA Isaac Sim/Isaac Lab, ROS/ROS2, and embedded AI.

Education

- **University of Illinois Chicago** — Chicago, IL
  Ph.D. in Electrical and Computer Engineering — *2024 – Present (Expected May 2029)*
- **University of Notre Dame** — Notre Dame, IN
  Ph.D. in Electrical Engineering — *2023–2024*
- **University of Pune** — Pune, India
  Bachelor of Engineering in Computer Engineering — *2015–2021*

Technical Skills

- **Robot Learning & Physical AI**: Deep Reinforcement Learning (PPO, SAC, DQN), Vision-Language-Action (VLA), Multimodal Robot Learning, Vision-Language Models (VLMs), Autonomous Navigation, Sim-to-Real, Motion/Path Planning, Multi-Agent Robotics, Sensor Fusion
- **Perception & Efficient AI**: Computer Vision, Monocular Depth Estimation, Object Detection (YOLO, SSD, FOMO), CNNs, Model Quantization (INT8), Knowledge Distillation, TinyML, Edge AI
- **Robotics & Simulation**: NVIDIA Isaac Sim, Isaac Lab, ROS/ROS2, Gazebo, MuJoCo, SOFA, MoveIt, RViz, URDF/USD, SolidWorks
- **Programming & ML Frameworks**: Python, C++, C, Embedded C, PyTorch, TensorFlow, OpenCV, Linux, Docker, Git/GitHub
- **Embedded Platforms & Sensors**: NVIDIA Jetson, Raspberry Pi, STM32, ESP32, Arduino, GAP8 (AI-deck), Intel RealSense, RPLIDAR, Robotiq FT 300 Force/Torque Sensor
- **Robotic Platforms**: Crazyflie 2.x (GAP8 AI-deck), Universal Robots UR5/UR5e, Clearpath Husky, Unitree Go2, Unitree G1

Experience

- **Research Assistant** — University of Illinois Chicago
  Conducting research under the supervision of Dr. Arman Roohi. — *Aug 2024 – Present*
  - **Crazyflie Isaac Framework (CIF) – Open-Source Robotics Simulation Framework**:
    Developed and open-sourced a modular NVIDIA Isaac Sim framework for Crazyflie research, integrating configurable controllers, sensors, environments, and autonomous algorithms for DRL, multi-drone, benchmarking, and sim-to-real experimentation. [GitHub]
  - **Vision-Language-Action (VLA) for Embodied Robotics — Ongoing**:
    Developing an application-specific VLA framework for resource-constrained robots, exploring DINO/SigLIP vision encoders, LoRA-adapted LLaMA models, knowledge distillation, imitation learning, and simulation-based RL for task-specific policy learning.
  - **Vision-Based Tiny-Drone Navigation**:
    Developed and deployed a fully onboard Crazyflie navigation stack integrating INT8 monocular depth, multi-range sensing, occupancy mapping, and dynamic path planning on GAP8.
  - **Edge Perception / Temporal Consistency**:
    Developed lightweight TinyML perception for resource-constrained edge platforms, including temporally consistent obstacle detection, embedded inference, BLE, and energy-aware sensing.
- **Research Assistant** — University of Notre Dame
  Conducted research under the supervision of Dr. Yasemin Ozkan Aydin. — *May 2023 – Aug 2024*
  - **Learning-Based Quadruped Locomotion**:
    Developed DQN-based locomotion for a salamander-inspired quadruped, integrating spinal–limb coordination and validating policies in simulation and on a physical Dynamixel/Raspberry Pi platform. (Demo).
  - **Autonomous Tunnel Burrowing System**: Developed and field-tested an autonomous excavation robot integrating depth perception, force/torque sensing, motion planning, and interchangeable tools. (Demo).
- **Robotics Software Engineer** — Kontrolmatik Technologies
  Motion Planning & Robot Control — *Jan 2022 – Sep 2022*
  - **Robot Motion Planning**:
    Developed C++/ROS motion-planning and control software for UR5e robots using inverse kinematics, collision checking, MoveIt, OMPL, and CHOMP.

- **Teaching Assistant** — University of Illinois Chicago
  Computer Architecture (Senior-Level Undergraduate Course) — *Jan 2025 – Present*
  - **Teaching & Mentorship**: Delivered a SimpleScalar tutorial, supported students in computer architecture topics, and graded assignments/exams with technical feedback and mentorship.

Selected Projects

- **OK-1: Custom Quadruped Robot**:
  Designed and built a custom quadruped robot with embedded electronics and locomotion control, implementing inverse kinematics and gait generation and validating control in ROS/Gazebo before physical deployment.
  (Demo) (Simulation) (GitHub)
- **18-DOF Humanoid Robot**:
  Designed and built an 18-DOF humanoid robot with custom mechanics, embedded control, and vision-based perception using Arduino, Raspberry Pi.
- **Custom FDM 3D Printer**:
  Designed and built a custom FDM 3D printer with embedded control, customized Marlin firmware, and Wi-Fi monitoring/control.
  (Demo)

Selected Publications

- **HORIZON: Lightweight Mapless Navigation for Resource-Constrained Micro Robots in Dynamic Environments.**
  O. Kurkutlu, R. Chien, M. Basit, and A. Roohi. *DroneSys*, 2026. **Accepted.**
- **Analyzing the Role of Spinal Joint Dynamics in the Movement of a Sprawling Robot.**
  M. Atasever, A. Nazaripouya, A. Okhovat, J. Nisbet, **O. Kurkutlu**, J. V. Deshmukh, and Y. Ozkan-Aydin. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- **LiMPNet: Lightweight Multi-Sensor Perception and DRL Navigation for Tiny Drones in Mapless Environments.**
  O. Kurkutlu and A. Roohi. *AAAI Fall Symposium Series*, 2025.
- **Lightweight Temporal Consistency for Grid-Based Obstacle Detection in Edge Devices.**
  O. Kurkutlu and A. Roohi. *IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion (ACSOS-C)*, 2025.
- **Evaluating Tools for Collective Soil Excavation: Challenges and Insights.**
  S. Even, **O. Kurkutlu**, and Y. Ozkan-Aydin. *International Symposium on Experimental Robotics (ISER)*, 2025.

Selected Research Presentations

- **IROS 2026 Late-Breaking Results**
  Lightweight 3D-Aware Navigation for Tiny Drones in Dynamic Mapless Environments, poster presentation, Sep. 2026.
- **DAC 2026 Young Fellows Program**
  Tiny Brain, Big World: Autonomous MCU-Friendly Navigation for Mapless Dynamic Environments, poster presentation, Design Automation Conference, 2026.
- **IROS 2023 Late-Breaking Results**
  Investigating the Impact of Spinal Joint Dynamics on a Sprawling Robot Using Deep Reinforcement Learning, poster presentation, 2023.

Awards & Honors

- **Sigma Xi** – Associate Member.
- **DAC 2026 Young Fellows Program** – Selected participant.